



Laboratory Work 4

Optimal control design using Bellman's dynamic programming

Lab Goal: *To design an optimal controller LTI plant using the bellman's dynamic programming*

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1. Introduction

In this laboratory work we will design the optimal controller for the LTI plant using the Bellman's dynamic programming to find the required controller parameters, and then compare the controller we made using the Riccati equation in the previous lab.

$$\dot{x} = Ax + Bu \rightarrow \text{Plant}$$

We will use the equations of optimal control law for a 2x2 system and the equation to synthesize the controller.

$$A = \begin{bmatrix} 0 & 1 \\ 4 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 2 \end{bmatrix}$$

While the Q matrix is

$$Q = \begin{bmatrix} 9 & 0 \\ 0 & 7 \end{bmatrix}, \quad R = 1$$

We need to solve for the parameters ψ that will be used for the control signal calculation,

$$u = \frac{1}{2r} ((2\psi_1 x_1 + \psi_{12} x_2) b_1 + (\psi_{12} x_1 + 2\psi_2 x_2) b_2)$$

```

a11=0;a12=1;a21=4;a22=-5;
b1=9; b2=2;
q1=9; q12=0; q2=7; r=1;
syms p11 p12 p22;
eqn1 = q1+2*a11*p11+a21*p12-(b1*p11/sqrt(r))^2-(b2*p12/(2*sqrt(r)))^2-b1*b2*p11*p12/r==0;
eqn2=q12+2*a12*p11+2*a21*p22+a11*p12+a22*p12-b1*b2*p12^2/(2*r)-b1^2*p11*p12/r-b2^2*p22*p12/r-2*b1*b2*p11*p22/r==0;
eqn3=q2+2*a22*p22+a12*p12-(b2*p22/sqrt(r))^2-(b1*p12/(2*sqrt(r)))^2-b1*b2*p22*p12/r==0;
sol=solve([eqn1, eqn2, eqn3], [p11, p22, p12], 'Real',true) ;
p1=double(sol.p11);
p2=double(sol.p22) ;
p_12=double(sol.p12) ;
for i=1
    P_eig=eig([p1(i) p_12(i)/2;p_12(i)/2 p2(i)]);
end

```

Figure 1. Calculation for Matrix P

For the verification of P, we first need to analyze the eigen values of P that it should be positive real values, at iteration 1 we have calculated the eigen values to be real positive while on other iteration it has a minimum of 1 negative value.

For further clarification we will compare the matrix P with the LQR,

```

P=[p1(1) p_12(1)/2;p_12(1)/2 p2(1)];
A=[a11 a12;a21 a22] ;B=[b1;b2] ;
Q=[q1 q12/2; q12/2 q2] ;
R=r;
[K,S]=lqr(A, B, Q, R);
disp(P)
disp(K)
disp(S)

```

Figure 2. Comparison with LQR

0.3352	-0.0260
-0.0260	0.6011
2.9651	0.9679
0.3352	-0.0260
-0.0260	0.6011

Figure 3. Comparison with LQR and bellman's solution

2. Modeling

We will now calculate the controller using the calculated parameters and then model the system on simulink to verify the output signals and states along with the cost function.

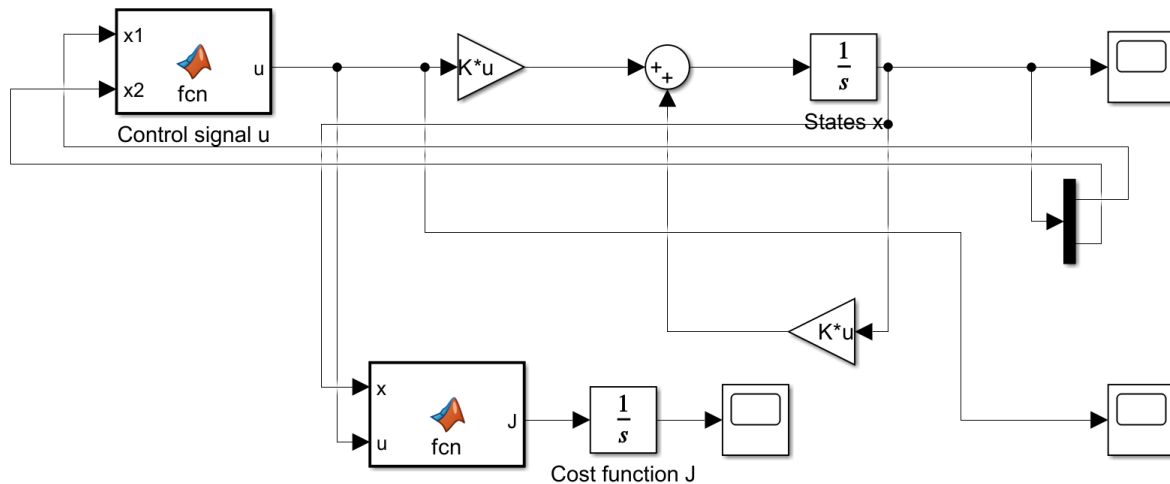


Figure 4. System Model

```
1 function u = fcn(x1,x2)
2
3 p1= 0.3352;
4 p2= 0.6011;
5 p_12 = -0.0260;
6 r=1;
7 b1 = 9;
8 b2 = 2;
9
10 u = -1/(2*r)*((2*x1*p1+p_12*x2)*b1 +(p_12*x1+2*p2*x2)*b2);
```

Figure 5. Controller calculation

3. Output plots

We will now simulate the system on the given initial conditions as $x(0) = [1; 0]^T$

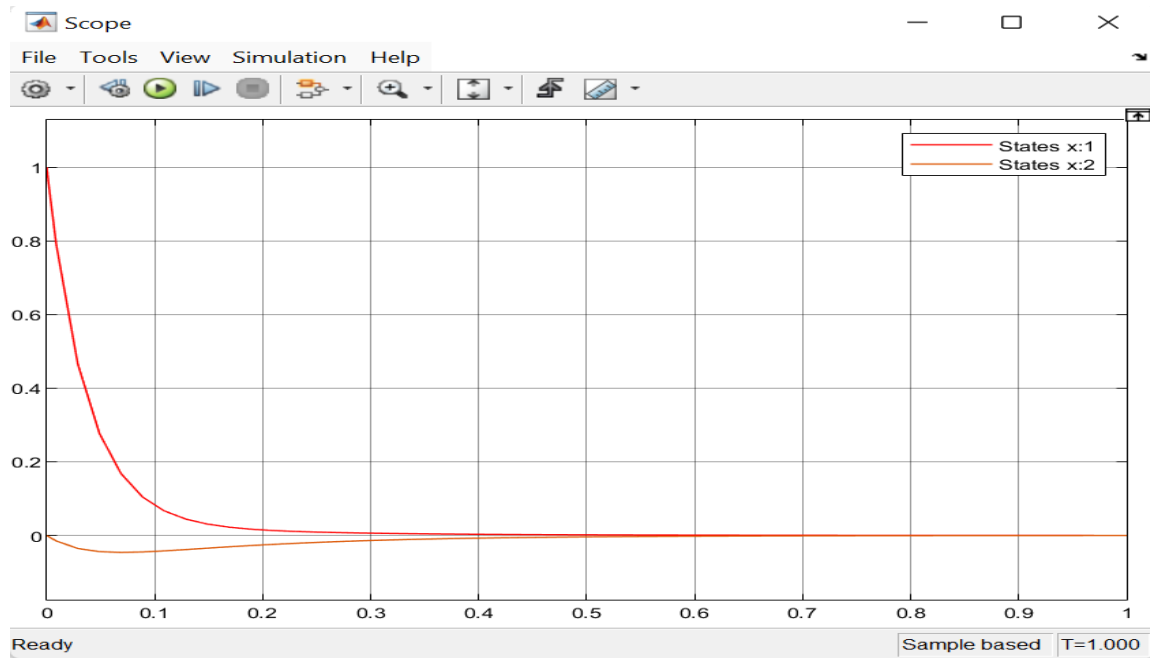


Figure 6. States x1 and x2

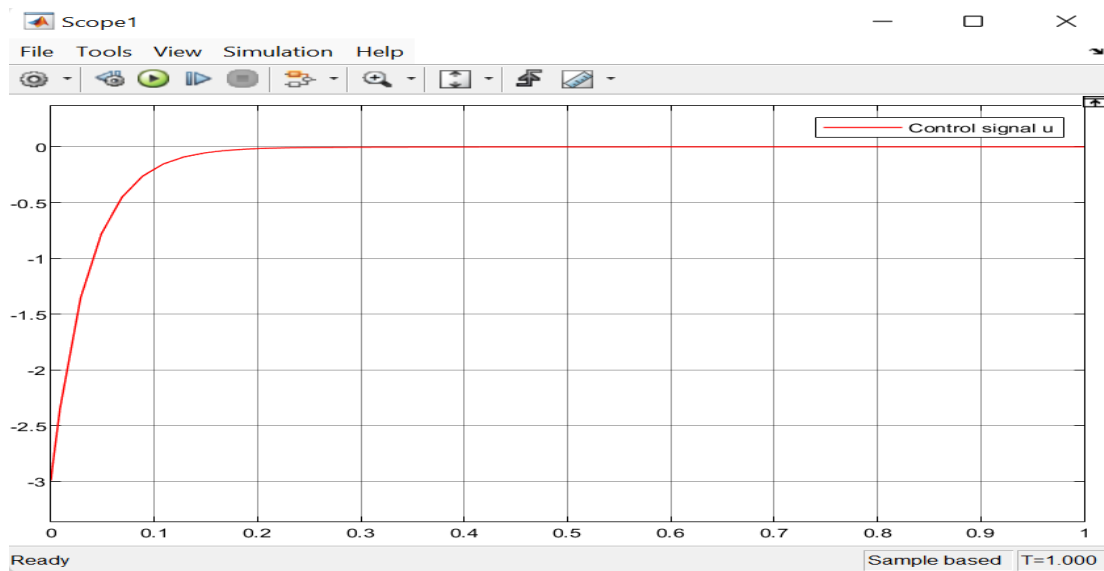


Figure 7. Control signal u

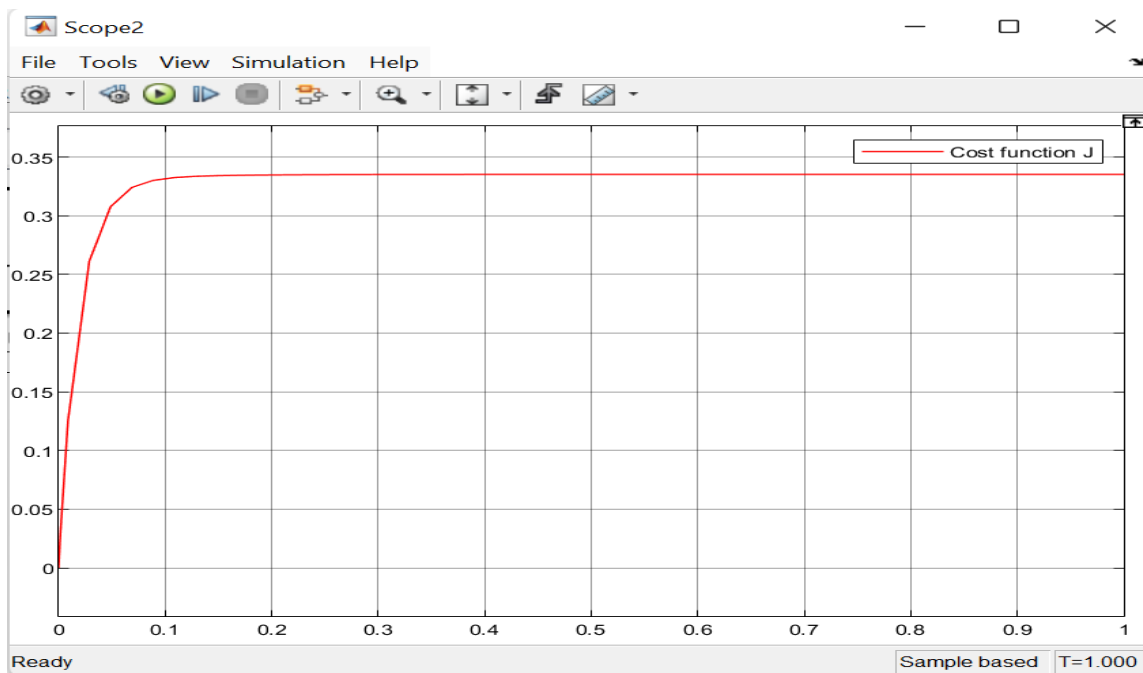


Figure 8. Cost function J

Since we have used the same parameters as the previous work, the optimal controller is calculated correctly as the cost function and output states coincides with the Riccati equation results, we will now change the controller parameters other than the optimal parameters and then check the obtained results.

```

1  function u = fcn(x1,x2)
2
3      p1= 0.3352+1;
4      p2= 0.6011+1;
5      p_12 = -0.0260+1;
6      r=1;
7      b1 = 9;
8      b2 = 2;
9
10     u = -1/(2*r)*((2*x1*p1+p_12*x2)*b1 +(p_12*x1+2*p2*x2)*b2);|

```

Figure 9. Controller with changed parameters

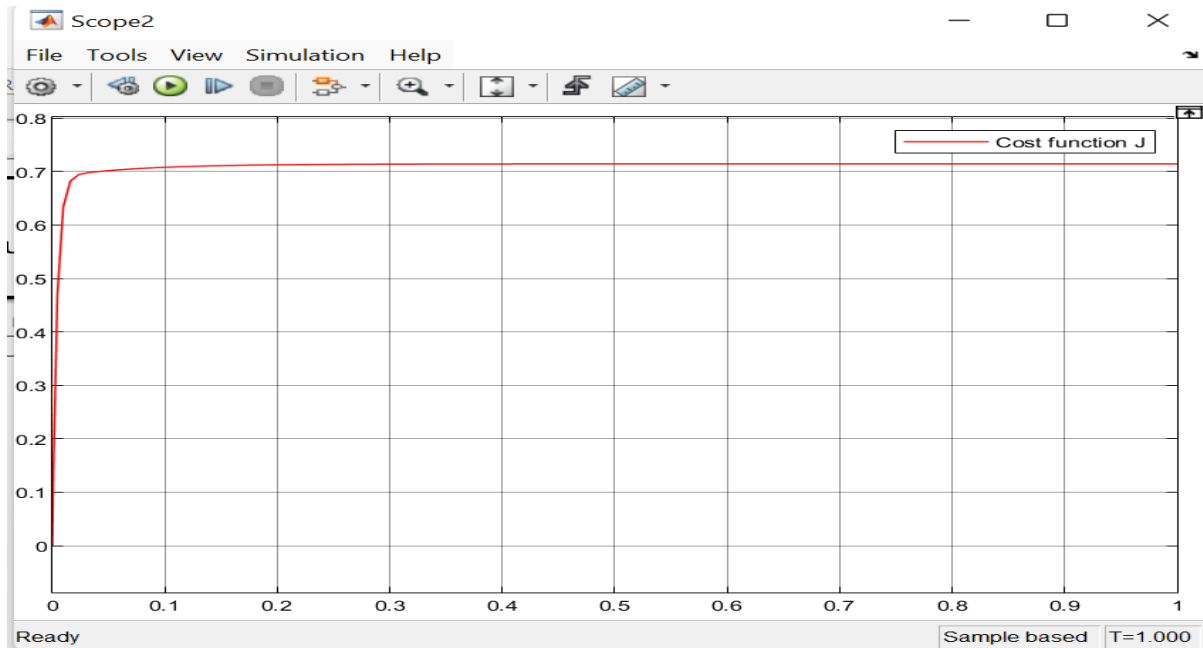


Figure 10. Cost function J on changed parameters

4. Conclusion

In this laboratory we have used the bellman's dynamic equation to successfully calculate the controller parameters for the optimal control and the least cost function

We have observed that by changing the optimal controller parameters the cost function of the system goes higher which means that the function will take more cost to drive the states to zero.